
MarginOptimizer

B2B Dynamic Alternative Flight & Yield Management
System

STRANDS AGENT AUTOMATED PNR ANALYSIS

CONFIDENTIAL · 2026

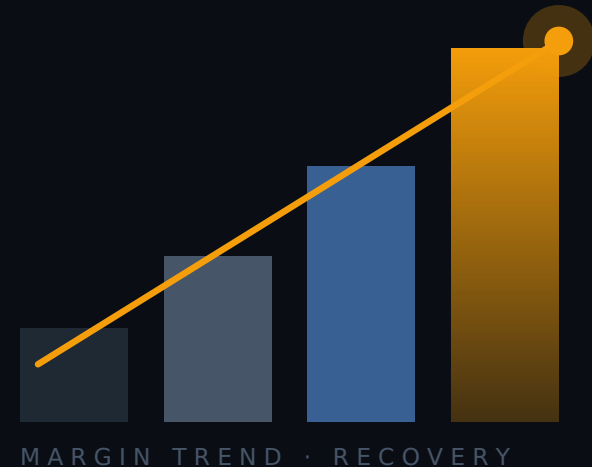


TABLE OF CONTENTS

What This Document Covers

01

The Cost Factor

Extracting invisible profit from waiting tickets.

02

The Solution Lifecycle

How the autonomous agent operates at 02:00 (Europe/Berlin).

03

Infrastructure & Scaling

Processing thousands of items without triggering API bans (SQS).

04

The Approval Loop (HITL)

Slack webhooks and operator-side security.

05

Development Roadmap

From local validation to cloud dashboard deployment.

THE COST FACTOR

Creating Money From Inefficiency

The Scenario: A €1,000 flight is locked in at booking. Two months later an identical seat — same baggage, same time window — drops to €800. No human can scan 10,000 PNRs nightly, so that €200 margin slips away.



ROI: ~50,000 yearly bookings × 2% profitable-hit rate × €40 average saving = **≥ €40,000 / year** in pure recovered margin.

SYSTEM ARCHITECTURE

AWS Serverless High-Throughput Engine



SERVICE	FUNCTION IN THE ARCHITECTURE
AWS EventBridge	Nightly cron at 02:00 Europe/Berlin kicking the ingest Lambda
Amazon SQS	Queues 10,000 bookings with a DLQ, shielding Traffics from bursts
Lambda (Worker)	One Strands + Haiku 4.5 agent per PNR, max 10 concurrent
API Gateway	Receives Slack approval webhooks, HMAC-verified before routing
Aurora Postgres	Writes every approved swap to <code>yield_events</code> ; feeds QuickSight

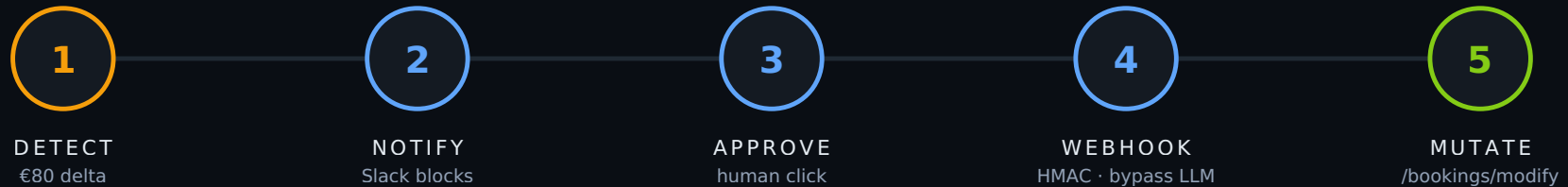
STRANDS TOOLING

The Margin Finders

- `use_traffics` ▶ `/offers/{code}/alternativeFlights`
- Pings the Traffics spine to query identical alternative planes.
- `calculator_tool` ▶ deterministic (New – Old) delta
- Prevents LLM mathematical hallucinations — arithmetic is never model-generated.
- `slack_tool` ▶ operations channel `#yield-ops`
- Dispatches interactive UI blocks (Approve/Reject) showing the found margin.

THE APPROVAL LOOP

Zero Unwanted Actions



The agent runs fully autonomous up to step 2. At step 4 the Webhook **bypasses the LLM entirely** — deterministic code (and only deterministic code) ever touches production bookings. One locked path in, one locked path out.

INVESTMENT & ROADMAP

Execution Strategy



01 · VALIDATION	0-1 mo	Prove the prompt logic isolates profit perfectly, locally.
02 · CLOUD FARM	1-2 mo	Deploy AWS SQS. Tune rate-limit handling for 429s.
03 · OPERATIONAL	2-3 mo	Slack webhooks + direct mutation scripts.
04 · BIG DATA	4+ mo	Amazon QuickSight "Yield Dashboards" for C-level.

MarginOptimizer

Autonomous, Invisible, Highly Profitable

CONFIDENTIAL & PROPRIETARY B2B YIELD
OPTIMIZATION AI.